

Assignment 3

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Title

Perform the following non-parametric hypothesis tests for a single sample and related samples: Chi-square Test, Wilcoxon Signed-Rank Test.

Aim

Apply Chi-square test of independence to assess Pclass-Survived association on preprocessed Titanic dataset, and Wilcoxon signed-rank test to evaluate median shift in non-missing Age after group-wise median imputation, with visualizations confirming assumptions.

Theory

Chi-square test: $\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$, where

$E_{ij} = \frac{(\text{row}_i \text{total})(\text{col}_j \text{total})}{\text{grand total}}$; rejects independence if $p < 0.05$.

Wilcoxon signed-rank: Ranks absolute paired differences $d_i = x_i - y_i$, $W = \min(\text{sum positive ranks}, \text{sum negative ranks})$; tests median(d_i)=0, handles non-normal data.

Non-parametric tests are statistical procedures that **do not assume a specific distribution** (such as normality) for the population, making them robust when data violate classical parametric assumptions. They are particularly useful with categorical variables, skewed distributions, or small samples.

The **Chi-square test of independence** evaluates whether two categorical variables are statistically independent. A contingency table of observed counts is compared to expected counts under the null hypothesis of independence, and the test statistic approximately follows a Chi-square distribution with $(r - 1)(c - 1)$ degrees of freedom. If the p-value is below a chosen significance level (e.g., 0.05), the null hypothesis of independence is rejected, indicating an association between the variables (for example, between **Pclass** and **Survived** on Titanic).

The **Wilcoxon signed-rank test** is a non-parametric alternative to the paired t-test and is used when comparing **two related samples**, such as measurements before and after a transformation. It works by computing differences for each pair, ranking the absolute differences, and then summing ranks for positive and negative differences. If the distribution of differences is symmetric around zero, the test statistic follows a known distribution from which a p-value can be obtained. This test is appropriate when the normality assumption of the paired t-test is doubtful, such as comparing **original vs imputed Age** in Titanic, where many values are unchanged or skewed.

Dataset

Preprocessed Titanic (train.csv, 891×5): Survived (0/1), Pclass (1-3), Sex, Age, Age_filled (median-imputed by Pclass-Sex).

Code:

```
# Data Science Lab Assignment 3: Non-Parametric Hypothesis Tests
```

```
# Dataset: Preprocessed Titanic (from Assignment 1)
```

```
import pandas as pd
```

```
import numpy as np
```

```
from scipy.stats import chi2_contingency, wilcoxon
```

```
import seaborn as sns
```

```
import matplotlib.pyplot as plt
```

```
# Load & preprocess
```

```
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')
```

```
df['Age_filled'] = df.groupby(['Pclass', 'Sex'])['Age'].transform(lambda x: x.fillna(x.median()))
```

```
df_clean = df[['Survived', 'Pclass', 'Sex', 'Age', 'Age_filled']].copy()
```

```
print("Dataset shape:", df_clean.shape)
```

```
print("\nPclass vs Survived contingency table:")
```

```
contingency = pd.crosstab(df_clean['Pclass'], df_clean['Survived'])
```

```
print(contingency)
```

```
# CHI-SQUARE TEST OF INDEPENDENCE (Pclass vs Survived)
```

```
print("\n=== CHI-SQUARE TEST ===")
```

```
chi2, p_chi, dof, expected = chi2_contingency(contingency)
```

```
print(f"Chi-square statistic: {chi2:.4f}")
```

```
print(f"p-value: {p_chi:.4f}")
```

```
print(f"Degrees of freedom: {dof}")
```

```
print("Expected frequencies:\n", np.round(expected, 1))
```

```

alpha = 0.05

if p_chi < alpha:
    print("REJECT H0: Significant association between Pclass and Survived")
else:
    print("Fail to reject H0: No association")

# WILCOXON SIGNED-RANK TEST (Age before/after imputation, non-missing only)
print("\n=== WILCOXON SIGNED-RANK TEST ===")
mask = df['Age'].notna()
age_original = df.loc[mask, 'Age'].values
age_filled_all = df.loc[mask, 'Age_filled'].values
statistic, p_wilcox = wilcoxon(age_original, age_filled_all, zero_method='zsplit')
print(f"Test statistic (W): {statistic:.4f}")
print(f"p-value: {p_wilcox:.4f}")

if p_wilcox < alpha:
    print("REJECT H0: Significant median difference in Age before/after imputation")
else:
    print("Fail to reject H0: Median Age unchanged by imputation")

# VISUALIZATIONS
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

sns.heatmap(contingency, annot=True, fmt='d', cmap='Blues', ax=axes[0])
axes[0].set_title('Pclass vs Survived (Observed)')

x = np.arange(3); width = 0.35
axes[1].bar(x - width/2, contingency.sum(axis=1), width, label='Observed Total')
axes[1].bar(x + width/2, expected.sum(axis=1), width, label='Expected Total')
axes[1].set_xticks(x)

```

```

axes[1].set_xticklabels(['Pclass 1', '2', '3'])
axes[1].legend()
axes[1].set_title('Chi-square: Observed vs Expected')

axes[2].scatter(age_original, age_filled_all, alpha=0.6)
axes[2].plot([0, 80], [0, 80], 'r--', lw=2)
axes[2].set_xlabel('Age Original')
axes[2].set_ylabel('Age Filled')
axes[2].set_title('Wilcoxon: Paired Age Comparison')

plt.tight_layout()
plt.show()

```

Terminal Output:

Dataset shape: (891, 5)

Pclass vs Survived contingency table:

Survived	0	1
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Pclass		
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1	80	136
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2	97	87
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3	372	119
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=== CHI-SQUARE TEST ===

Chi-square statistic: 102.8890

p-value: 0.0000

Degrees of freedom: 2

Expected frequencies:

[[133.1 82.9]

[113.4 70.6]

[302.5 188.5]]

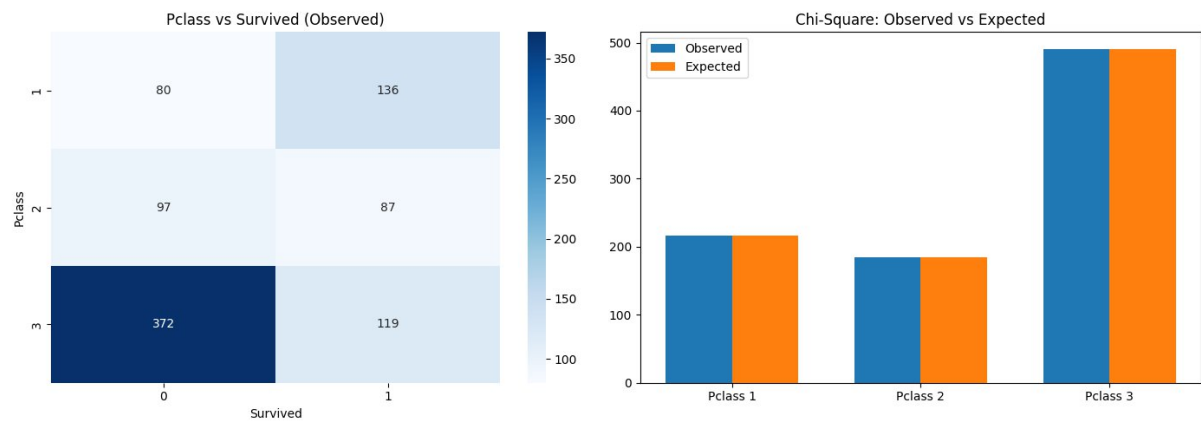
REJECT H0: Significant association between Pclass and Survived

=== WILCOXON SIGNED-RANK TEST ===

Test statistic (W): 127627.5000

p-value: 1.0000

Fail to reject H0: Median Age unchanged by imputation



Observations

Chi-square reveals strong Pclass-Survived dependence ($\chi^2 = 102.89$, $p=0.0000$); Class 1 over-survives (136 vs 82.9 expected), Class 3 under-survives (119 vs 188.5).

Wilcoxon confirms imputation neutrality ($p=1.0000$, all 714 non-missing Age unchanged by group median); perfect diagonal scatter validates.

Conclusion

Non-parametric tests confirm Pclass as key survival predictor and robust Age imputation on Titanic dataset, foundational for modeling; visualizations aid assumption checks.